

MTH 166

Lecture-35

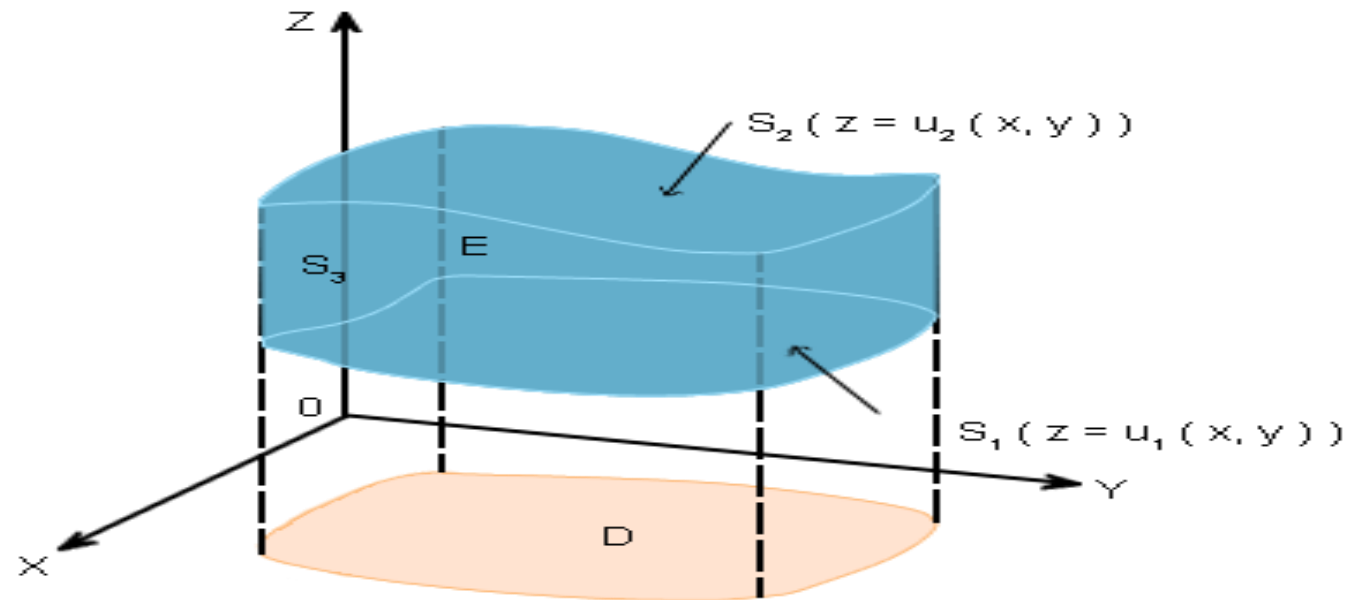
Surface Area and Surface Integral

Area Element and Surface Area

1. If the projection of the surface $z = f(x, y)$ is taken on xy -plane.

$$\text{Area element, } dA = \sqrt{1 + f_x^2 + f_y^2}$$

$$\text{Surface Area, } A = \iint_R dA = \iint_R \sqrt{1 + f_x^2 + f_y^2} dx dy$$



2. If the projection of the surface $x = g(y, z)$ is taken on yz -plane.

$$\text{Area element, } dA = \sqrt{1 + g_y^2 + g_z^2}$$

$$\text{Surface Area, } A = \iint_R dA = \iint_R \sqrt{1 + g_y^2 + g_z^2} dydz$$

3. If the projection of the surface $y = h(z, x)$ is taken on zx -plane.

$$\text{Area element, } dA = \sqrt{1 + h_z^2 + h_x^2}$$

$$\text{Surface Area, } A = \iint_R dA = \iint_R \sqrt{1 + h_z^2 + h_x^2} dzdx$$

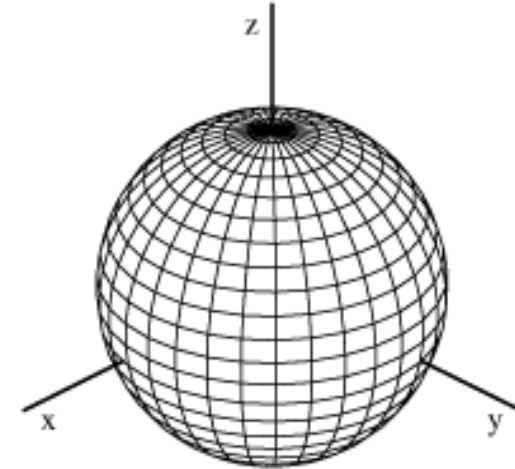
Important Questions from MCQ Point of View:

Problem 1. Find the surface area of $x^2 + y^2 + z^2 = a^2$

Solution: Since it is sphere of radius a

So, Surface area of sphere, $S = 4\pi r^2$

$\Rightarrow S = 4\pi(a)^2$ **Answer.**

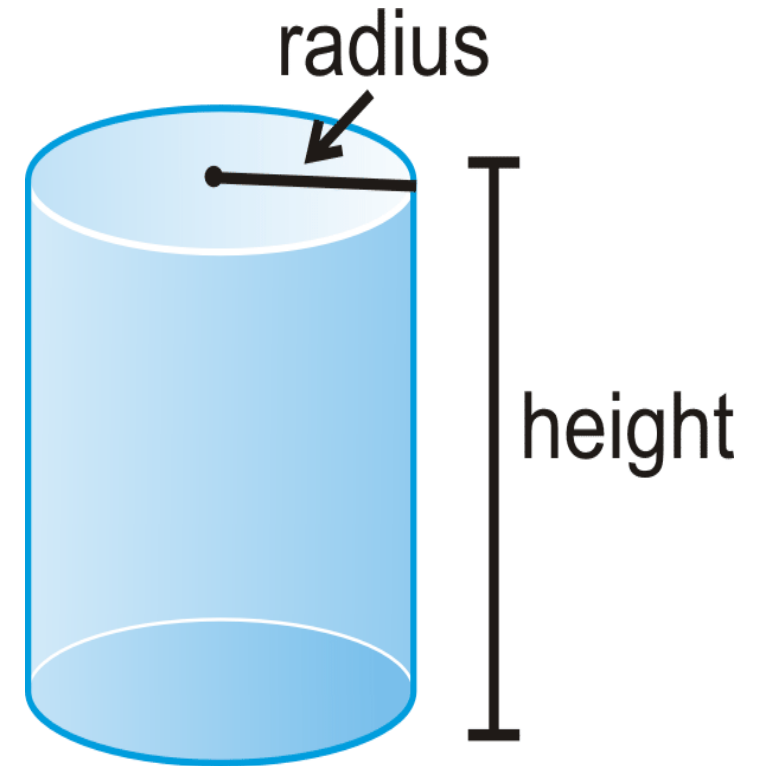


Problem 2. Find the surface area of $x^2 + y^2 = 16, 0 \leq z \leq 2$

Solution: Since it is cylinder of radius 4 and height 2.

So, Surface area of cylinder, $S = 2\pi rh$

$\Rightarrow S = 2\pi(4)(2) = 16\pi$ **Answer.**



Problem 3. Find the surface area of $z^2 = x^2 + y^2, 0 \leq z \leq 4$

Solution: Since it is a cone of height $h = 4$ and radius of topmost part $r = 4$.

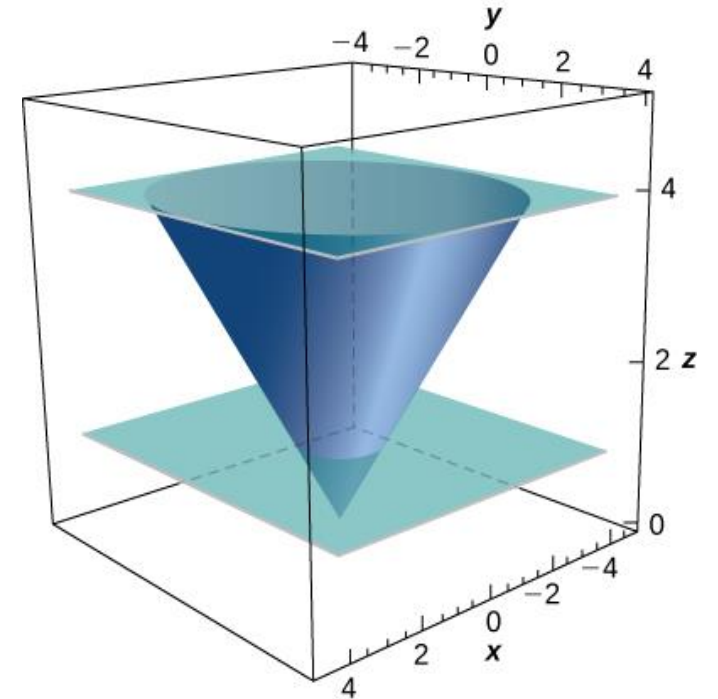
Here slanting height, $l = \sqrt{h^2 + r^2}$

$$\Rightarrow l = \sqrt{(4)^2 + (4)^2} = 4\sqrt{2}$$

So, Surface area of cone, $S = \pi r l$

$$\Rightarrow S = \pi(4)(4\sqrt{2})$$

$$\Rightarrow S = 16\sqrt{2}\pi \text{ Answer.}$$



Polling quiz

If the orthogonal projection of the surface $z = f(x, y)$ is taken on the xy plane, the element surface area dA is given by:

(a) $\sqrt{f_x^2 + f_y^2}$

(b) $\sqrt{1 + f_x^2 + f_y^2}$

(c) $\sqrt{1 + f_y^2 + f_z^2}$

(d) $\sqrt{1 + f_z^2 + f_x^2}$

Surface Integral

The expression for surface integral is: $I = \iint_S f(x, y, z) dA$

Where dA is the area element and it can be determined from one of the following ways:

If projection is taken on xy-plane ($z=0$ plane), then: $dA = \frac{dx dy}{\hat{n} \cdot \hat{k}}$

If projection is taken on yz-plane ($x=0$ plane), then: $dA = \frac{dy dz}{\hat{n} \cdot \hat{i}}$

If projection is taken on zx-plane ($y=0$ plane), then: $dA = \frac{dz dx}{\hat{n} \cdot \hat{j}}$

Problem 4. Evaluate the surface integral $\iint_S f(x, y, z) dA$ where $f = 6xyz$ and S is the portion of the plane $x + y + z = 1$ in the first octant.

Solution: Let $g = x + y + z = 1$

$$\Rightarrow g_x = 1, g_y = 1, g_z = 1$$

$$\text{Now, } \vec{\nabla} g = g_x \hat{i} + g_y \hat{j} + g_z \hat{k} = \hat{i} + \hat{j} + \hat{k}$$

$$\hat{n} = \frac{\vec{\nabla} g}{|\vec{\nabla} g|} = \frac{(\hat{i} + \hat{j} + \hat{k})}{\sqrt{1+1+1}}$$

$$\Rightarrow \hat{n} = \frac{(\hat{i} + \hat{j} + \hat{k})}{\sqrt{3}}$$

Let projection of Surface S be taken on xy-plane:

$$z = 1 - x - y$$

$$dA = \frac{dxdy}{\hat{n} \cdot \hat{k}} = \frac{dxdy}{\frac{(\hat{i} + \hat{j} + \hat{k})}{\sqrt{3}} \cdot \hat{k}}$$

$$\Rightarrow dA = \frac{dxdy}{\frac{1}{\sqrt{3}}}$$

For limits: On xy-plane($z = 0$)

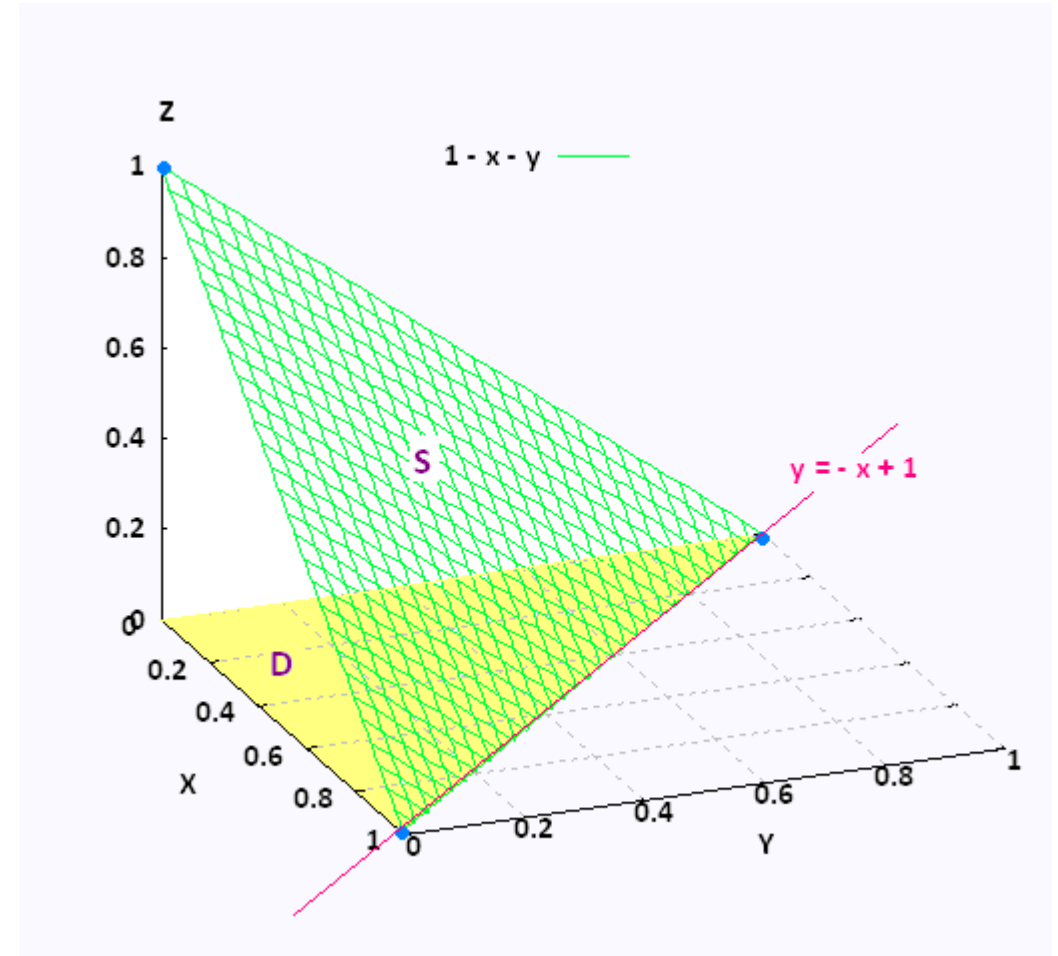
$$\text{So, } x + y + 0 = 1 \Rightarrow y = (1 - x)$$

$$0 \leq y \leq (1 - x)$$

On x-axis ($y = 0, z = 0$)

$$\text{So, } x + 0 + 0 = 1 \Rightarrow x = 1$$

$$0 \leq x \leq 1$$



So, required surface integral is: $I = \iint_S f(x, y, z) dA$

$$\begin{aligned}\Rightarrow I &= \iint_S (6xyz) \frac{dx dy}{\frac{1}{\sqrt{3}}} \\ &= 6\sqrt{3} \int_{x=0}^1 \int_{y=0}^{1-x} xy(1-x-y) dx dy \\ &= 6\sqrt{3} \int_{x=0}^1 \int_{y=0}^{1-x} (xy - x^2y - xy^2) dy dx\end{aligned}$$

On simplification, we get:

$$I = \frac{\sqrt{3}}{20} \textbf{Answer.}$$

Polling quiz

Which of the following options is **Not** true for element surface area dA :

(a) $dA = \frac{dydz}{\hat{n} \cdot \hat{n}}$

(b) $dA = \frac{dydz}{\hat{n} \cdot \hat{i}}$

(c) $dA = \frac{dzdx}{\hat{n} \cdot \hat{j}}$

(d) $dA = \frac{dxdy}{\hat{n} \cdot \hat{k}}$



THANK
you